

## **Project Name:** Predicting Alzheimer's Disease Severity from Blood Biomarkers

**Background:** Alzheimer's disease is a progressive neurodegenerative disorder in which biological changes often begin before major cognitive symptoms are visible. While cognitive evaluations such as CDR-SB, MMSE, MoCA, and FAQ are commonly used to assess impairment, they can be difficult and time-intensive to conduct. Blood-based biomarkers, by contrast, are easier to obtain and may offer a more accessible and scalable way to capture Alzheimer's-related pathology. Our project tests whether these biomarkers can predict cognitive evaluation scores, with the broader goal of improving assessment of disease severity.

**Goal:** Use blood-based biomarkers to predict clinical severity measures in Alzheimer's disease, including CDR-SB for global severity, FAQ for functional impairment, and MMSE/MoCA for cognitive performance, and explore how these predictions can support stage classification.

### **Dataset:** ADNI dataset

- Blood biomarkers: A $\beta$ 40, A $\beta$ 42, A $\beta$ 42/A $\beta$ 40 ratio, AlzPath p-tau217, GFAP, Janssen p-tau217, NfL, np-tau217, p-tau181, p-tau217, and p-tau217 ratio
- Additional predictor: APOE genotype
- After cleaning: 907 rows for CDR, 977 for FAQ, 392 for MoCA, and 970 for MMSE

### **How to Start (Tools Needed):**

1. Use separate **LASSO** regression models for each cognitive outcome to select the most informative biomarker features with **sklearn**.
2. Use the selected biomarkers as inputs for **random forest** models to predict cognitive test scores such as CDR, MMSE, FAQ, or MoCA.
3. Combine predicted cognitive scores with diagnosis information to generate an overall assessment of disease severity.

### **Stretch Goal(s):**

1. Build a more detailed severity report integrating biomarker signals across functional, cognitive, and global measures
2. Compare which biomarkers are most useful for different types of outcomes

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## Example:

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| Project:                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Background (paragraph)       | The reference human genome sequence is inarguably the most important and widely used resource in the fields of human genetics and genomics. However, its use introduces innate biases towards the sequence variants present in the reference. A collection of many human genomes, organized into the human pangenome reference, can reduce the number of missing variants for downstream analysis. However, the effect of introducing missing variants has not been quantified in many common analysis including GWAS, Functional Analysis, and Pathogenicity. Nor is it well understood how these effects vary by genome coverage or by population group.                                 |
| Goal                         | Can we replicate results from HPRC initial paper determining that we can recover more variants using the pangenome.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Dataset                      | <ul style="list-style-type: none"><li>- Human Pangenome Reference subset down to 2 regions. (1 highly variable and 1 without much variation)</li><li>- 1000 Genomes reads over those regions with simulated phenotype information</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Suggest way to start (tools) | <ol style="list-style-type: none"><li>1. Align reads<ol style="list-style-type: none"><li>a. to pangenome using <b>graphaligner</b> or <b>giraffe</b></li><li>b. to linear reference genome with <b>bwa-mem</b> or other</li></ol></li><li>2. Call variants<ol style="list-style-type: none"><li>a. from pangenome set using <b>vg</b></li><li>b. from linear set using <b>deep variant</b> or <b>broad tool</b></li></ol></li><li>3. Downstream analysis<ol style="list-style-type: none"><li>a. Use <b>pandas</b> to determine patterns in types of variants</li><li>b. Use <b>plink2</b> to run GWAS using both sets of variants</li><li>c. Pathogenicity something</li></ol></li></ol> |
| Stretch Goal                 | <ol style="list-style-type: none"><li>1. Determine how these effects change by coverage and population group</li><li>2. Quantify the effect the additional variants have on downstream analysis including GWAS and determine if patterns in pathogenicity or variant type emerge.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                |